Question	Answer	Marks	AOs		Guidance
8	EITHER			**DR**	
	Equation of the form $y = k(x+1)(x-2)$	**M1**	**1.1a**	Allow with $k = 1$ and without $y =$	Ignore = 0 if seen
	(0, –4) on curve so $k = 2$	**M1**	**3.1a**	Attempt to find $k \neq 1$	
		A1	**1.1b**	All correct	
	OR				
	Equation of the form $y = ax^2 + bx + c$				
	(0, -4) on curve $c = -4$	**(M1)**		Uses one point to form an equation	Allow for c = -4 seen
	(-1, 0) on the curve $0 = a - b - 4$ (2, 0) on the curve $0 = 4a - 2b - 4$	**(M1)**		Uses both other points and attempts to solve simultaneous equations	
	Solving simultaneous equations $a = 2,\ b = -2$	**(A1)**		All correct	
	BOTH				
	Area = $\int_{-1}^{2}\left(2x^2 - 2x - 4\right)\mathrm{d}x$				
	$\left[\frac{2x^3}{3} - x^2 - 4x\right]_{-1}^{2}$	**M1**	**1.1a**	Integration – allow without limits – condone one error	
		A1	**1.1b**	FT their quadratic	
	$\left(\frac{2\times 2^3}{3} - 2^2 - 4\times 2\right) - \left(\frac{2\times(-1)^3}{3} - (-1)^2 - 4\times(-1)\right)$	**M1**	**1.1a**	Substitution of limits clearly seen Complete argument leading to exact answer.	
	$= -\frac{20}{3} - \frac{7}{3} = -9$	**A1**	**2.1**	Allow for 9 if there is an argument to explain the change of sign even if -9 not seen.	
	Area is 9 below the x-axis.	**E1**	**2.4**	Must give modulus and explain the change of sign. FT if their definite integral is negative.	"Area must be positive" is not sufficient explanation.
		[8]			